DCF-Driven Personalized Semantic Field Architecture

Formal Foundations for Dynamic Embedding, Personalized Field Resonance, Family-Word Retrieval, and Intent-Guided Response Composition

Kevin T.N
Vietnam
keedavinci@gmail.com

March 18, 2026

Abstract

This document formalizes a missing architectural element for a personalized semantic intelligence system: a Dynamic Context Field (DCF) framework that separates semantic understanding from personal resonance. The architecture is built around five distinct but coupled processes: contextual semantic interpretation, field resonance against a personalized internal profile, family-word retrieval from semantic memory, intent inference, and response composition. Unlike conventional large language models in which latent space emerges implicitly through large-scale parameter optimization, the present framework treats latent semantic structure as an explicitly constructed field whose coordinates are generated through context-sensitive resonance. A token is therefore not assigned a fixed meaning vector once and for all; instead, it is represented as a distribution over semantic prototypes, whose weights vary dynamically under contextual activation.

The central claim of this document is that understanding what a speaker means and determining how an artificial system should respond are not the same computation. First, language must be mapped into a shared semantic latent space. Second, that representation must be evaluated against the personalized field state of the AI, derived from its history of interaction, reinforcement, and internal preference structure. Only then should the system retrieve semantically related family-word clusters and reorganize them under inferred communicative intent.

The framework is formalized using explicit definitions for tokens, semantic prototypes, base fields, context planes, contextual embeddings, field profiles, resonance operators, family-word clusters, and intent functions. Core equations are provided for prototype weighting, semantic embedding construction, field resonance scoring, family retrieval, and intent-guided compositional response. The result is not a full implementation manual but a formal specification layer from which implementation can later be synthesized. In this sense, the document is intended as a principled substrate specification for a more interpretable and personalized alternative to purely parameter-heavy language architectures.

Contents

1 Problem Statement **4**

2 System Objective **4**

3 Core Definitions **4**
3.1 Token . . . 4
3.2 Semantic Prototype . . . 5
3.3 Base Field . . . 5
3.4 Context . . . 5
3.5 Anchor Set . . . 5
3.6 Context Plane . . . 5
3.7 Semantic Resonance . . . 5
3.8 Personal Field Profile . . . 5
3.9 Field Resonance . . . 6
3.10 Family-Word Cluster . . . 6
3.11 Intent . . . 6

4 Foundational Principles **6**
4.1 Separation of Semantic Understanding and Personal Resonance . . . 6
4.2 Non-Fixed Meaning of Tokens . . . 6
4.3 Context as Weight Configuration . . . 6
4.4 Constructed Latent Semantic Space . . . 6
4.5 Response Mediation . . . 6

5 Mathematical Objects **6**
5.1 Token Prototypes . . . 7
5.2 Input Utterance Representation . . . 7
5.3 Personal Field State . . . 7

6 Context Plane Construction **7**
6.1 Anchor Extraction . . . 7
6.2 Anchor Embedding Matrix . . . 8
6.3 Context Plane . . . 8

7 Semantic Resonance and Dynamic Embedding **8**
7.1 Prototype Resonance . . . 8
7.2 Prototype Weights . . . 9
7.3 Dynamic Token Embedding . . . 9
7.4 Utterance-Level Semantic Representation . . . 9

8 Personalized Field Resonance **9**
8.1 Semantic-to-Field Projection . . . 9
8.2 Normalized Field Resonance Vector . . . 10
8.3 Interpretation . . . 10

9 Family-Word Retrieval **10**
9.1 Family Memory Store . . . 10

9.2 Family Retrieval Score . . . 11
9.3 Retrieved Family Set . . . 11
9.4 Interpretation . . . 11

10 Intent Inference **11**
10.1 Intent Set . . . 11
10.2 Intent Score . . . 11
10.3 Intent Distribution . . . 11
10.4 Interpretation . . . 12

11 Response Composition **12**
11.1 Compositional Inputs . . . 12
11.2 Response Plan . . . 12
11.3 Response Composition Operator . . . 12
11.4 Interpretation . . . 13

12 Full System Pipeline **13**

13 Training Regimes **13**
13.1 Semantic Field Training . . . 13
13.2 Personal Field Resonance Training . . . 14
13.3 Family-Word Memory Training . . . 14
13.4 Intent Training . . . 14

14 Invariants **15**
14.1 Semantic Invariants . . . 15
14.2 Field Invariants . . . 15
14.3 Retrieval Invariants . . . 15
14.4 Intent Invariants . . . 15
14.5 Composition Invariants . . . 15

15 Worked Example: The Token “cá” **15**

16 Contrast with Conventional LLM Architectures **16**

17 Naming and Positioning **17**

18 Conclusion **17**

1. Problem Statement

Conventional language models conflate several distinct computational tasks into one large probabilistic mechanism. They learn token embeddings, contextual transformations, preference regularities, discourse patterns, and response generation within a single parameterized stack. This produces impressive fluency, but at the cost of interpretability. It becomes difficult to distinguish what the system understands from how it reacts, what belongs to shared semantics from what belongs to internal preference, and what arises from the speaker's intent versus what arises from the model's learned stylistic bias.

The architecture formalized here proposes an alternative decomposition. It begins from the claim that the following computations should be separated:

1. semantic interpretation of the user's input;
2. personalized resonance of that semantic content with the AI's internal field profile;
3. retrieval of semantically related linguistic material appropriate to the activated field state;
4. inference of the speaker's communicative intent;
5. final organization of response material into a coherent utterance.

Core thesis. Semantic understanding and personalized response are distinct layers. A language system should first determine *what is being said*, then determine *what resonates internally*, then determine *what family of language is available*, and only then determine *how to answer*.

This document gives the formal layer for that decomposition.

2. System Objective

The objective is to define a field-based language architecture in which:

- a shared semantic latent space supports contextual interpretation;
- a personalized field profile supports individualized resonance;
- field activation retrieves semantically related family-word clusters;
- speaker intent reorganizes retrieved material into appropriate discourse.

The architecture therefore aims to formalize a system in which language generation is not a direct next-token extrapolation from raw decoding, but the end result of a structured chain of semantic and field operations.

3. Core Definitions

3.1. Token

Definition 3.1 (Token)**.** *A* token *is any elementary linguistic unit admitted by the system for semantic processing. A token may be a word, phrase, entity, idiomatic fragment, or any structured symbolic unit. We denote tokens by T.*

3.2. Semantic Prototype

Definition 3.2 (Semantic Prototype)**.** *A* semantic prototype *of a token T is a context-specific meaning profile $C_i(T)$ representing one possible semantic mode of that token. Each prototype is associated with an anchor set, a field-weight profile, and a prototype vector in semantic space.*

For a token T, let its prototype family be

$$\mathscr{C}(T) = \{C_1(T), C_2(T), \ldots, C_{n_T}(T)\}.$$

3.3. Base Field

Definition 3.3 (Base Field)**.** *A* base field *is a semantic axis or semantic force dimension within the global semantic substrate. The total field set is*

$$\mathscr{F} = \{F_1, F_2, \ldots, F_m\}.$$

A base field is not itself a full meaning category; it is a foundational coordinate direction in semantic space.

3.4. Context

Definition 3.4 (Context)**.** *A* context *is a weight configuration over a multidimensional semantic field. It is not an external label attached to a token, but a distribution that activates one or more semantic prototypes and base-field directions.*

3.5. Anchor Set

Definition 3.5 (Anchor Set)**.** *An* anchor set *$\mathscr{A}$ is a finite collection of words, entities, concepts, or discourse markers used to stabilize a local context plane for interpretation.*

3.6. Context Plane

Definition 3.6 (Context Plane)**.** *A* context plane *is a local semantic projection surface generated from an anchor set and a local linguistic neighborhood. It defines the active interpretive geometry under which token resonance is computed.*

3.7. Semantic Resonance

Definition 3.7 (Semantic Resonance)**.** Semantic resonance *is the degree to which a token or prototype aligns with an active context plane. It is a context-dependent compatibility score rather than a fixed similarity value.*

3.8. Personal Field Profile

Definition 3.8 (Personal Field Profile)**.** *A* personal field profile *of an AI is a vector-valued internal state*

$$\mathbf{f}^{(AI)} = [f_1, \ldots, f_m] \in \mathbb{R}^m$$

augmented by memory, preference, and reinforcement statistics over the base fields. It represents how semantic input resonates with that particular AI instance.

3.9. Field Resonance

Definition 3.9 (Field Resonance)**.** Field resonance *is the response of the AI's personal field profile to a semantic representation. It yields a personalized activation vector over the base fields.*

3.10. Family-Word Cluster

Definition 3.10 (Family-Word Cluster)**.** *A* family-word cluster *is a semantically organized group of lexical and phrasal material linked by field affinity, semantic proximity, discourse function, and stylistic compatibility.*

3.11. Intent

Definition 3.11 (Intent)**.** *The* intent *of an utterance is the communicative action inferred from the speaker's input, such as requesting information, seeking confirmation, challenging a claim, sharing emotion, or prompting reflection.*

4. Foundational Principles

4.1. Separation of Semantic Understanding and Personal Resonance

Principle 4.1. *Understanding what a speaker means and determining how a given AI should respond are distinct computational processes.*

The first process should be relatively universal across agents. The second should be personalized.

4.2. Non-Fixed Meaning of Tokens

Principle 4.2. *A token does not possess a single fixed meaning vector. Rather, it is represented as a distribution over semantic prototypes, weighted dynamically by context.*

4.3. Context as Weight Configuration

Principle 4.3. *A context is not a label but a weight distribution over semantic fields and prototype activations.*

4.4. Constructed Latent Semantic Space

Principle 4.4. *Latent semantic space should be explicitly constructed through context-sensitive field operations, rather than treated only as an emergent side effect of large-scale statistical parameter optimization.*

4.5. Response Mediation

Principle 4.5. *A response should not be generated directly from raw semantic decoding. It should be mediated through personalized field resonance, family-word retrieval, and intent-sensitive reorganization.*

5. Mathematical Objects

We now define the core objects used in the formalization.

5.1. Token Prototypes

For each token T, let

$$\mathscr{C}(T) = \{C_1, \ldots, C_n\}$$

be its prototype family. Each prototype C_i carries:

- an anchor set $\mathscr{A}_i$;
- a prototype vector $\mathbf{v}_i \in \mathbb{R}^d$;
- a base-field weight vector $\mathbf{b}_i \in \mathbb{R}^m$;
- metadata such as usage statistics, discourse tags, and lexical neighborhood.

5.2. Input Utterance Representation

Let the user input be a sequence

$$X = (T_1, T_2, \ldots, T_\ell).$$

From X, the system extracts a local anchor set

$$\mathscr{A}(X) = \{a_1, \ldots, a_k\}$$

and constructs a context plane Π_X.

5.3. Personal Field State

The personalized field state is represented by:

$$\mathbf{f}^{(AI)} = [f_1, \ldots, f_m] \in \mathbb{R}^m.$$

More generally, each field F_j may carry a tuple

$$F_j^{(AI)} = (f_j, h_j, p_j, d_j),$$

where:

- f_j: current activation weight,
- h_j: resonance history,
- p_j: preference or affinity coefficient,
- d_j: memory density or reinforcement depth.

6. Context Plane Construction

6.1. Anchor Extraction

Given input X, define an anchor extraction operator:

$$\mathscr{A}(X) = \text{ExtractAnchors}(X).$$

This may depend on:

- token salience,
- syntactic role,
- lexical neighborhood,
- discourse markers,
- prior semantic clustering.

6.2. Anchor Embedding Matrix

Let each anchor a_j have a preliminary semantic vector $\mathbf{u}_j \in \mathbb{R}^d$. Define the anchor matrix:

$$U_X = \begin{bmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \\ \vdots \\ \mathbf{u}_k^\top \end{bmatrix} \in \mathbb{R}^{k \times d}.$$

Let the anchor centroid be

$$\bar{\mathbf{u}}_X = \frac{1}{k} \sum_{j=1}^{k} \mathbf{u}_j.$$

Centered anchor matrix:

$$\widetilde{U}_X = U_X - \mathbf{1}\bar{\mathbf{u}}_X^\top.$$

6.3. Context Plane

A simple formalization of the context plane is the subspace spanned by the dominant directions of $\widetilde{U}_X$. Let

$$\Pi_X = \mathrm{Span}\{\mathbf{e}_1, \ldots, \mathbf{e}_r\},$$

where $\mathbf{e}_1, \ldots, \mathbf{e}_r$ are the leading singular vectors of $\widetilde{U}_X$.

Remark 6.1. *This is one admissible formalization. More general constructions may include graph-induced anchor geometry, weighted manifold projections, or nonlinear context charts.*

7. Semantic Resonance and Dynamic Embedding

7.1. Prototype Resonance

For a token T and one of its prototypes C_i, define the semantic resonance score:

$$R_i(T \mid X) = \mathrm{Res}\big(C_i, \Pi_X, \mathscr{A}(X)\big).$$

One possible operational form is:

$$R_i(T \mid X) = \alpha\, \mathrm{sim}(\mathbf{v}_i, \bar{\mathbf{u}}_X) + \beta\, \mathrm{ProjScore}(\mathbf{v}_i, \Pi_X) + \gamma\, \mathrm{AnchorOverlap}(C_i, \mathscr{A}(X)),$$

where $\alpha, \beta, \gamma \geq 0$ are coefficients.

Here:

$$\begin{aligned}
\mathrm{sim}(\mathbf{v}_i, \bar{\mathbf{u}}_X) &= \frac{\mathbf{v}_i \cdot \bar{\mathbf{u}}_X}{\|\mathbf{v}_i\| \, \|\bar{\mathbf{u}}_X\|}, \\
\mathrm{ProjScore}(\mathbf{v}_i, \Pi_X) &= \frac{\|\mathrm{Proj}_{\Pi_X}(\mathbf{v}_i)\|}{\|\mathbf{v}_i\|}, \\
\mathrm{AnchorOverlap}(C_i, \mathscr{A}(X)) &= \frac{|\mathscr{A}_i \cap \mathscr{A}(X)|}{|\mathscr{A}_i \cup \mathscr{A}(X)|}.
\end{aligned}$$

7.2. Prototype Weights

Prototype weights are obtained through softmax normalization:

$$w_i(T \mid X) = \frac{\exp(R_i(T \mid X)/\tau)}{\sum_{k=1}^{n_T} \exp(R_k(T \mid X)/\tau)},$$

where $\tau > 0$ is a temperature parameter.

Thus:

$$\sum_{i=1}^{n_T} w_i(T \mid X) = 1.$$

7.3. Dynamic Token Embedding

The contextual embedding of token T under input X is:

$$\mathbf{e}(T \mid X) = \sum_{i=1}^{n_T} w_i(T \mid X) \, \mathbf{v}_i.$$

This is a dynamic embedding because the weights $w_i(T \mid X)$ depend on context.

7.4. Utterance-Level Semantic Representation

For an input sequence $X = (T_1, \ldots, T_\ell)$, define the utterance semantic representation as:

$$\mathbf{s}(X) = \mathrm{Agg}\Big(\mathbf{e}(T_1 \mid X), \ldots, \mathbf{e}(T_\ell \mid X)\Big),$$

where Agg may be a weighted average, attention-like aggregation, graph pooling, or composition operator.

A simple form is:

$$\mathbf{s}(X) = \sum_{j=1}^{\ell} \lambda_j \mathbf{e}(T_j \mid X), \qquad \sum_{j=1}^{\ell} \lambda_j = 1,$$

with salience weights λ_j.

8. Personalized Field Resonance

8.1. Semantic-to-Field Projection

Let each base field F_j have a field vector $\mathbf{q}_j \in \mathbb{R}^d$ and AI-specific coefficients p_j, d_j. Define the raw field resonance:

$$\rho_j(X) = \mathrm{FieldRes}_j\big(\mathbf{s}(X), F_j^{(AI)}\big).$$

One admissible form is:

$$\rho_j(X) = \eta_1 \operatorname{sim}(\mathbf{s}(X), \mathbf{q}_j) + \eta_2 p_j + \eta_3 d_j + \eta_4 \operatorname{HistMatch}_j(X),$$

where $\eta_1, \eta_2, \eta_3, \eta_4 \geq 0$.

8.2. Normalized Field Resonance Vector

Define the personalized field resonance distribution:

$$r_j(X) = \frac{\exp(\rho_j(X)/\kappa)}{\sum_{h=1}^{m} \exp(\rho_h(X)/\kappa)}, \qquad j = 1, \ldots, m,$$

where $\kappa > 0$ is a field temperature.

Thus:

$$\mathbf{r}(X) = [r_1(X), \ldots, r_m(X)] \in \mathbb{R}^m, \qquad \sum_{j=1}^{m} r_j(X) = 1.$$

Definition 8.1 (Personalized Field Resonance Vector)**.** *The vector* $\mathbf{r}(X)$ *is called the* personalized field resonance vector *of utterance* X.

8.3. Interpretation

The semantic embedding $\mathbf{s}(X)$ captures what the utterance means in the shared semantic space. The field vector $\mathbf{r}(X)$ captures how that semantic content resonates with the internal field structure of the AI.

Separation of roles.

$\mathbf{s}(X)$ answers: what does the utterance mean?

$\mathbf{r}(X)$ answers: how does that meaning resonate with this AI?

9. Family-Word Retrieval

9.1. Family Memory Store

Let the semantic memory store contain family clusters:

$$\mathscr{W} = \{W_1, \ldots, W_N\}.$$

Each family cluster W_u carries:

- a cluster vector $\mathbf{g}_u \in \mathbb{R}^d$,
- a field tag vector $\mathbf{z}_u \in \mathbb{R}^m$,
- lexical items and phrases,
- discourse function metadata,
- style or tone metadata.

9.2. Family Retrieval Score

Given semantic representation $\mathbf{s}(X)$ and field resonance $\mathbf{r}(X)$, define the score of family cluster W_u:

$$\phi_u(X) = \mu_1 \operatorname{sim}(\mathbf{s}(X), \mathbf{g}_u) + \mu_2 \operatorname{sim}(\mathbf{r}(X), \mathbf{z}_u) + \mu_3 \operatorname{ContextFit}(W_u, X) + \mu_4 \operatorname{StyleFit}(W_u, X),$$

where $\mu_1, \mu_2, \mu_3, \mu_4 \geq 0$.

9.3. Retrieved Family Set

Let

$$\mathscr{R}(X) = \operatorname{TopK}\{\phi_u(X)\}_{u=1}^{N}$$

denote the top-K retrieved family clusters.

Equivalently, define a soft family distribution:

$$\pi_u(X) = \frac{\exp(\phi_u(X)/\sigma)}{\sum_{v=1}^{N} \exp(\phi_v(X)/\sigma)}.$$

Definition 9.1 (Family-Word Retrieval)**.** *The* family-word retrieval step *maps input X to a ranked or weighted family-cluster set*

$$\mathscr{R}(X) = \operatorname{RetrieveFamily}\big(\mathbf{s}(X), \mathbf{r}(X), X\big).$$

9.4. Interpretation

This layer determines *what linguistic material becomes available for response* after semantic understanding and field resonance have already occurred.

10. Intent Inference

10.1. Intent Set

Let the intent class set be:

$$\mathscr{I} = \{I_1, \ldots, I_L\}.$$

Example classes may include information request, clarification request, confirmation seeking, challenge, emotional sharing, instruction, or reflective prompting.

10.2. Intent Score

Define intent score for class I_ℓ:

$$\psi_\ell(X) = \nu_1 \operatorname{DiscMarker}(X, I_\ell) + \nu_2 \operatorname{SemCompat}(\mathbf{s}(X), I_\ell) + \nu_3 \operatorname{FieldCompat}(\mathbf{r}(X), I_\ell) + \nu_4 \operatorname{FamilyCompat}(\mathscr{R}(X), I_\ell),$$

where $\nu_1, \nu_2, \nu_3, \nu_4 \geq 0$.

10.3. Intent Distribution

$$\omega_\ell(X) = \frac{\exp(\psi_\ell(X))}{\sum_{h=1}^{L} \exp(\psi_h(X))}.$$

Thus:

$$\boldsymbol{\omega}(X) = [\omega_1(X), \ldots, \omega_L(X)].$$

The predicted intent is:

$$I^*(X) = \arg\max_{I_\ell \in \mathscr{I}} \omega_\ell(X).$$

Definition 10.1 (Intent Inference)**.** *The* intent inference layer *maps*

$$(X, \mathbf{s}(X), \mathbf{r}(X), \mathscr{R}(X)) \mapsto \boldsymbol{\omega}(X)$$

and selects an inferred intent $I^*(X)$.

10.4. Interpretation

A speaker may use semantically similar language under different intents. Therefore meaning and intent must not be collapsed into one variable.

11. Response Composition

11.1. Compositional Inputs

The response layer receives:

$$\Big(\mathbf{s}(X), \mathbf{r}(X), \mathscr{R}(X), I^*(X), P\Big),$$

where P is a persona or style policy.

11.2. Response Plan

Define a discourse plan variable:

$$D(X) = \text{Plan}\Big(\mathbf{s}(X), \mathbf{r}(X), \mathscr{R}(X), I^*(X), P\Big).$$

This plan may encode:

- abstraction level,
- tone,
- degree of directness,
- ordering of retrieved family material,
- answer length,
- explanatory vs reflective stance.

11.3. Response Composition Operator

Define the final response:

$$Y(X) = \text{Compose}\Big(\mathscr{R}(X), I^*(X), \mathbf{r}(X), P, D(X)\Big).$$

Proposition 11.1 (Mediated Response Generation)**.** *In this architecture, the response is not a direct function of raw token decoding alone, but a mediated composition:*

$$X \mapsto \mathbf{s}(X) \mapsto \mathbf{r}(X) \mapsto \mathscr{R}(X) \mapsto I^*(X) \mapsto Y(X).$$

11.4. Interpretation

This layer reorganizes available language material according to the inferred communicative purpose of the speaker.

12. Full System Pipeline

The full runtime pipeline can now be written formally as:

$$X \longrightarrow \Pi_X \longrightarrow \mathbf{s}(X) \longrightarrow \mathbf{r}(X) \longrightarrow \mathscr{R}(X) \longrightarrow I^*(X) \longrightarrow Y(X).$$

Expanded form:

$$\begin{aligned}
\mathscr{A}(X) &= \text{ExtractAnchors}(X),\\
\Pi_X &= \text{BuildContextPlane}(\mathscr{A}(X), X),\\
\mathbf{s}(X) &= \text{SemanticMap}(X, \Pi_X),\\
\mathbf{r}(X) &= \text{FieldResponse}(\mathbf{s}(X), \mathbf{f}^{(AI)}),\\
\mathscr{R}(X) &= \text{RetrieveFamily}(\mathbf{s}(X), \mathbf{r}(X), X),\\
I^*(X) &= \text{InferIntent}(X, \mathbf{s}(X), \mathbf{r}(X), \mathscr{R}(X)),\\
Y(X) &= \text{Compose}(\mathscr{R}(X), I^*(X), \mathbf{r}(X), P).
\end{aligned}$$

Architectural distinction. The semantic layer is the shared interpretive layer. The field layer is the personalized resonance layer. The family layer is the language-material activation layer. The intent layer is the communicative-action layer. The composition layer is the discourse generation layer.

13. Training Regimes

13.1. Semantic Field Training

The semantic field layer should be trained or constructed from corpora, ontologies, lexical relations, contextual examples, and manually seeded prototypes.

Objectives include:

- stabilizing prototype vectors $\mathbf{v}_i$,
- learning or calibrating anchor associations,
- refining context-plane geometry,
- improving contextual resonance scoring.

One possible training objective is:

$$\mathscr{L}_{sem} = \lambda_{ctx}\mathscr{L}_{ctx} + \lambda_{proto}\mathscr{L}_{proto} + \lambda_{align}\mathscr{L}_{align},$$

where:

- $\mathscr{L}_{ctx}$ penalizes wrong contextual prototype activations,
- $\mathscr{L}_{proto}$ regularizes prototype consistency,
- $\mathscr{L}_{align}$ aligns semantic structure with anchor geometry.

13.2. Personal Field Resonance Training

The personal field profile is trained from user-specific interaction data:

- conversation history,
- accepted or rejected responses,
- preference traces,
- long-term reinforcement patterns.

Let feedback be $q_t \in \mathbb{R}$ at interaction step t. Then a simple update for field j is:

$$f_j^{(t+1)} = (1-\alpha) f_j^{(t)} + \alpha\, q_t\, r_j(X_t),$$

where $r_j(X_t)$ is the field resonance produced by the current utterance.

This makes personalization accumulate over interactions.

13.3. Family-Word Memory Training

Family clusters can be learned or curated using:

- co-occurrence structure,
- discourse usage patterns,
- style clustering,
- field-conditioned lexical grouping.

One possible objective is:

$$\mathscr{L}_{fam} = \lambda_{sem}\mathscr{L}_{semantic_cluster} + \lambda_{field}\mathscr{L}_{field_cluster} + \lambda_{disc}\mathscr{L}_{discourse_cluster}.$$

13.4. Intent Training

Intent inference can be trained on dialogue acts, annotated intents, discourse markers, and personalized conversational data:

$$\mathscr{L}_{intent} = -\sum_{\ell=1}^{L} y_\ell \log \omega_\ell(X),$$

where y_ℓ is the one-hot target distribution.

14. Invariants

14.1. Semantic Invariants

- Every token must map to at least one semantic prototype.
- Prototype weights for a token under context must sum to one:

$$\sum_i w_i(T \mid X) = 1.$$

- Every utterance must produce a semantic embedding $\mathbf{s}(X)$.

14.2. Field Invariants

- Every semantic embedding must produce a normalized field resonance vector:

$$\sum_{j=1}^{m} r_j(X) = 1.$$

- Personalization should modify field response without destroying the shared semantic layer.

14.3. Retrieval Invariants

- Family retrieval must depend on both semantic fit and field fit.
- Retrieval must not reduce to raw token frequency.

14.4. Intent Invariants

- Intent must remain distinct from semantic content.
- Semantically similar utterances may carry different intents.

14.5. Composition Invariants

- The response must be coherent with inferred intent.
- The response must be consistent with field resonance.
- The response must be drawn from reorganized family material, not from unfiltered raw semantic memory.

15. Worked Example: The Token "cá"

Consider the Vietnamese token $T =$ "cá". Suppose its prototype family is:

$$\mathscr{C}(T) = \{C_1, C_2, C_3, C_4, C_5, C_6\},$$

with:

- C_1: fish as aquatic animal,

- C_2: fish as food ingredient,
- C_3: fish as prepared dish,
- C_4: fish as aquaculture resource,
- C_5: fish as cultural symbol,
- C_6: fish as ecological entity.

Now take the utterance:

$$X = \text{“Tôi ăn cá chiên”}.$$

Suppose the prototype resonance scores are:

$$R(T \mid X) = [1.2,\ 3.5,\ 2.8,\ 0.5,\ 0.1,\ 0.3].$$

After softmax normalization:

$$[w_1, \ldots, w_6] = [0.05,\ 0.56,\ 0.31,\ 0.04,\ 0.02,\ 0.02].$$

Thus the dynamic token embedding is:

$$\mathbf{e}(T \mid X) = 0.05\mathbf{v}_1 + 0.56\mathbf{v}_2 + 0.31\mathbf{v}_3 + 0.04\mathbf{v}_4 + 0.02\mathbf{v}_5 + 0.02\mathbf{v}_6.$$

Suppose the utterance-level semantic representation is then mapped to an AI-specific field resonance:

$$\mathbf{r}(X) = [0.08,\ 0.61,\ 0.17,\ 0.07,\ 0.04,\ 0.03].$$

This means that although the shared semantic understanding identifies the utterance primarily as food-related, the personalized AI field state further emphasizes specific active fields based on its own internal profile.

Family-word retrieval may then return clusters such as:

$$\mathscr{R}(X) = \{\text{“món ăn” cluster, “nguyên liu” cluster, “hng v” cluster}\}.$$

If intent inference predicts:

$$I^*(X) = \text{conversational sharing},$$

then the response composition layer may construct a reply around everyday food discourse rather than biological explanation.

Remark 15.1. *This example shows the central separation: the utterance is first understood semantically, then personalized through field resonance, then expanded through family retrieval, and only afterward composed as discourse.*

16. Contrast with Conventional LLM Architectures

A conventional LLM typically learns a large latent space through parameter optimization over massive corpora. Token embeddings, contextual disambiguation, discourse habits, and preference effects are

mixed into one large model. In contrast, the present framework separates these layers.

Aspect	Conventional LLM	DCF Field Architecture
Latent space origin	Emergent from training	Explicitly constructed through semantic fields and context resonance
Token meaning	Implicit contextual transformation	Distribution over prototypes with explicit dynamic weights
Personalization	Often shallow or external	Internalized as personalized field resonance
Response generation	Direct token prediction stack	Mediated by family retrieval and intent reorganization
Interpretability	Low	Higher, due to explicit field and prototype structure
Semantic vs personal layer	Entangled	Explicitly separated

17. Naming and Positioning

A suitable formal name for this missing project element is:

DCF-Driven Personalized Semantic Field Architecture

Alternative names include:

- Resonance-Based Dynamic Embedding Architecture,
- Personalized Semantic Field Resonance Model,
- Field-Conditioned Semantic Cognition Layer.

If the immediate goal is a concrete training application, then the implementation-facing name may be:

DCF Field Weight Trainer

or

Semantic Field Weight Studio

18. Conclusion

This document formalized a missing architectural component: a field-based semantic substrate that separates shared understanding from personalized resonance. The formalization introduced five core stages: contextual semantic mapping, AI-specific field resonance, family-word retrieval, intent inference, and response composition. Together, these stages define an alternative to opaque parameter-heavy language generation, one in which meaning is represented as dynamic contextual weighting over semantic prototypes, and personalized response emerges from explicit field interaction rather than hidden parameter entanglement.

The key conceptual move is simple but decisive. A language system should not respond directly from raw semantic decoding. It should first determine what the speaker means in a shared semantic space, then determine how that meaning resonates with the AI's personalized internal field, then retrieve the relevant

family of language material, and finally reorganize that material according to inferred communicative intent. Once stated in this form, the system is no longer a generic next-token predictor. It becomes a layered semantic-response architecture.

The present text is intentionally formal rather than implementational. Its purpose is to provide clear definitions, equations, invariants, and decomposition principles so that later code generation or application-layer design can proceed from a coherent substrate. In that sense, this document serves as a formal specification layer for a broader project whose missing element is precisely this: a principled mechanism for dynamic semantic weighting, personalized resonance, and intent-guided response construction.

Appendix A: Compact Formal Summary

For input utterance X, the architecture is defined by:

$$X \to \Pi_X \to \mathbf{s}(X) \to \mathbf{r}(X) \to \mathscr{R}(X) \to I^*(X) \to Y(X).$$

Where:

$$\begin{aligned}
\Pi_X &= \text{BuildContextPlane}(\text{ExtractAnchors}(X), X), \\
\mathbf{s}(X) &= \text{SemanticMap}(X, \Pi_X), \\
\mathbf{r}(X) &= \text{FieldResponse}(\mathbf{s}(X), \mathbf{f}^{(AI)}), \\
\mathscr{R}(X) &= \text{RetrieveFamily}(\mathbf{s}(X), \mathbf{r}(X), X), \\
I^*(X) &= \text{InferIntent}(X, \mathbf{s}(X), \mathbf{r}(X), \mathscr{R}(X)), \\
Y(X) &= \text{Compose}(\mathscr{R}(X), I^*(X), \mathbf{r}(X), P).
\end{aligned}$$

Appendix B: Minimal Data Schemas

Prototype Record

```
Prototype
- prototype_id
- token_id
- label
- anchor_set
- prototype_vector
- base_field_vector
- discourse_tags
- usage_stats
```

Personal Field Record

```
FieldProfile
- field_id
- current_weight
- preference_score
- resonance_history
```

```
- memory_density
- update_rules
```

Family Cluster Record

```
FamilyCluster
- cluster_id
- field_tag_vector
- semantic_cluster_vector
- words[]
- phrases[]
- tone_profile
- discourse_functions[]
```

Intent Record

```
IntentProfile
- intent_id
- label
- discourse_markers
- semantic_patterns
- field_patterns
- confidence_rules
```